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EE9-CS2-1
EE9-FPN2-05

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE/EIE PART IV: MEng and ACGI

STABILITY AND CONTROL OF NON-LINEAR SYSTEMS

Wednesday, 5 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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1. Consider the following nonlinear time-invariant autonomous system:

$$\begin{aligned} \dot{x}_1 &= -4x_1^3 + x_1 + 4x_2 \\ \dot{x}_2 &= 4x_1^2 + 4x_2^2 - 1 \end{aligned}$$

with state $x = [x_1, x_2]'$ evolving in $\mathbb{R}^2$.

a) Justify existence and uniqueness of solutions. [2]

b) Find and sketch the nullclines of the vector-field (Hint: find the intersection with axes and slope when crossing 0). [2]

c) Find regions in $\mathbb{R}^2$ delimited by nullclines and identify the orientation of the vector-field in each one of them (Hint: there should be 4 of them). [2]

d) Find, if any, forward invariant regions among those identified in part c). [2]

e) Compute all equilibria of the system. [2]

f) Linearize the system around each equilibrium and discuss the local phase-portrait. [3]

g) Consider the set $R := \{x : x_1 \geq 1/2, 4x_2 \geq 4x_1^3 - x_1\}$; sketch R within the phase-plane and discuss the asymptotic behaviour of solutions with initial conditions in R. [4]

h) Sketch the global phase portrait, emphasizing equilibria and how their local portraits merge into each other. [3]

2. Consider the following second-order differential equation:

$$\ddot{y}(t) = -\sin^2(\dot{y}(t))\dot{y}(t) - y^3(t) + u(t),$$

where $y(t)$ and $u(t) \in \mathbb{R}$.

a) Find a state-space description of the system, with input u and output y. [3]

b) Compute, as a function of input u, the corresponding equilibrium of the system and denote it by $x_e(u)$. [2]

c) Consider the system dynamics for $u = 0$. By interpreting the equations as describing the evolution of a unit mass subject to an elastic (cubic) force and a nonlinear friction term, propose a candidate Lyapunov function $V(x)$ for assessing stability of the origin. (Hint: use mechanical energy as a candidate Lyapunov function) [2]

d) Show that $V(x)$ is positive definite and radially unbounded. [3]

e) Compute the derivative $\dot{V}(x)$, along solutions of the system (for $u = 0$), and discuss its sign definiteness. [2]

f) Argue, through a suitable stability criterion, that 0 is a globally asymptotically stable equilibrium for the unforced system. [3]

g) Consider next arbitrary constant values of u; for the corresponding equilibrium point, derived in item b), define shifted coordinates $\tilde{x} = x - x_e(u)$. Write the system's equations in $\tilde{x}$ coordinates. [2]

h) How would you modify the Lyapunov function $V(x)$ to prove global asymptotic stability of the equilibrium $\tilde{x} = 0$? (or equivalently of $x = x_e(u)$?). [3]

3. Let Σ denote the following bidimensional system affected by disturbances:

$$\begin{aligned} \dot{x}_1 &= -x_1 + x_2^3 \\ \dot{x}_2 &= -x_2^9 + a x_1^3 + d \end{aligned}$$

with state $x \in \mathbb{R}^2$ and input d in $\mathbb{R}$. Assume that $a \in \mathbb{R}$ is an unknown constant parameter.

a) We wish to interpret the system as a feedback interconnection of two lower dimensional subsystems, Σ_1 and Σ_2, with inputs d_1 and d_2, respectively. Write the open-loop equations of the individual subsystems. [4]

b) Show that Σ_1 is an ISS system and compute its input-to-state gain γ_1, from input d_1 to state x_1. [4]

c) Show that Σ_2 is an ISS system and compute its a dependent input-to-state gain γ_2, from input d_2 to state x_2 (Hint: you may let $d = 0$). [4]

d) Find for which values of a the feedback interconnection of Σ_1 and Σ_2 fulfils the small gain condition, and show that Σ is ISS for all such values of a. [4]

e) Prove that for $a = 1$ (and for $d = 0$) the origin is not locally asymptotically stable (Hint: find out the equilibria of the system). [4]

4. Consider the SISO nonlinear control system of equations:

$$\begin{aligned} \dot{x}_1 &= x_3 + \sin(x_2) \\ \dot{x}_2 &= -x_2^3 + x_1^3 \\ \dot{x}_3 &= \sin(x_3) + (2 + \cos(x_2))u \\ \\ y &= x_1 \end{aligned}$$

with state $x = [x_1, x_2, x_3]'$ in $\mathbb{R}^3$, input $u \in \mathbb{R}$ and output $y \in \mathbb{R}$.

a) Find the relative degree of the system and explicitly state if this is well-defined, locally or globally (justify your answer). [3]

b) Find an input-output linearising state-feedback control law. [3]

c) Design a feedback to stabilize at $y = 0$ the input-output linear model achieved in the previous question. [3]

d) Express the system equations in normal form and highlight the zero-dynamics. [5]

e) Discuss Input-to-State Stability of the zero-dynamics and argue how this might enable the global asymptotic stabilization of the system. [6]